

YANPENG YU

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EDUCATION

Yale University, New Haven, CT, USA

09/2021– 05/2026 (expected)

Ph.D. in Computer Science

Advisors: Profs. Anurag Khandelwal and Lin Zhong

Peking University, Beijing, China

09/2017 – 06/2021

B.S. in Computer Science

Exchange program: Stanford University, 2019 Summer

AREA OF EXPERTISE

- Computer systems, computer architectures, distributed systems, operating systems, memory disaggregation, cache coherence protocols, synchronization, networks, system and architecture for AI/ML/LLM
- C/C++, CUDA, Python, PyTorch, vLLM, Linux, RDMA

WORK EXPERIENCE

NVIDIA Corporation, Architecture Research Group, *Research Intern*, Santa Clara, CA

05/2025 – 08/2025

- Multi-GPU load-balancing for efficient expert-parallel mixture-of-expert (MoE) model serving.
- Reduced Qwen3 and DeepSeek-V3 decode latency by up to 22% and improved token throughput by up to 21% compared to DeepSeek’s EPLB.
- Paper in submission.

NVIDIA Corporation, Architecture Research Group, *Research Intern*, Santa Clara, CA

05/2024 – 08/2024

- High-performance and energy-efficient cache coherence protocols for CPU-GPU shared memory.
- Reduced CPU-GPU collaborative benchmarks’ running time by 24% and inter-PU communication traffic by 13% compared to existing heterogeneous cache coherence protocols.
- Paper accepted to ISCA ’25, won Distinguished Artifact Award; selected for IEEE Micro’s Top Picks in Computer Architecture in 2025; Techniques applied to NVIDIA Vera Rubin Platform.

SELECTED PUBLICATIONS AND PREPRINTS

- **Yanpeng Yu**, Seung-seob Lee, Lin Zhong, and Anurag Khandelwal. [GCS: Generalized Cache Coherence For Efficient Synchronization](#). In *Proceedings of the 20th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2026.
- **Yanpeng Yu***, Haiyue Ma*, (*equal contribution) Krish Agarwal, Nicolai Oswald, Qijing Huang, Hugo Linsenmaier, Chunhui Mei, Ritchie Zhao, Ritika Borkar, Bitu Rouhani, David Nellans, Ronny Krashinsky, Anurag Khandelwal. [Efficient MoE Serving in the Memory-Bound Regime: Balance Activated Experts, Not Tokens](#). *Under Review*, 2025.
- **Yanpeng Yu**, Nicolai Oswald, and Anurag Khandelwal. [CORD: Low-Latency, Bandwidth-Efficient and Scalable Release Consistency via Directory Ordering](#). In *Proceedings of the 52nd Annual International Symposium on Computer Architecture (ISCA)*, 2025. 🏆 **Distinguished Artifact Award**. Selected for 🏆 **IEEE Micro’s Top Picks in Computer Architecture in 2025**.
- Lei Zou, Fan Zhang, Yinnian Lin, and **Yanpeng Yu**. [An Efficient Data Structure for Dynamic Graph on GPUs](#). *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 2023.
- Seung-seob Lee, **Yanpeng Yu**, Yupeng Tang, Anurag Khandelwal, Lin Zhong, and Abhishek Bhattacharjee. [MIND: In-Network Memory Management for Disaggregated Data Centers](#). In *Proceedings of the ACM SIGOPS 28th Symposium on Operating Systems Principles (SOSP)*, 2021.

AWARDS

🏆 **Distinguished Artifact Award**, 52nd Annual International Symposium on Computer Architecture (ISCA) 2025

🏆 **IEEE Micro’s Top Picks in Computer Architecture in 2025** 2025

LEADERSHIP EXPERIENCE

Athena Student Leadership Council, Athena AI Institute, *Student Leader*

2024 – 2025

TEACHING EXPERIENCE

- Teaching Assistant, CPSC 438/538 Big Data Systems: Trends & Challenges, Yale University 2023 Fall
- Teaching Assistant, CPSC 437 Introduction to Database Systems, Yale University 2022 Fall